

Program-9: Steps to Execute

Step 1: Create Project Directory and start minikube

```
mkdir Program-9  
cd Program-9
```

Step 2: Create namespaces.yaml

```
nano namespaces.yaml
```



```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/prog9  
GNU nano 7.2 namespace.yaml  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod
```

Step 3: Create pods.yaml

```
nano pods.yaml
```

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/prog9
GNU nano 7.2 pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

Step 4: Apply Namespace File and Pods File

```
kubectl apply -f namespaces.yaml
```

```
kubectl apply -f pods.yaml
```

Step 5: Verify Pods by Namespace

```
kubectl get pods -n dev
```

```
kubectl get pods -n prod
```

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/prog9
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$ kubectl apply -f namespace.yaml
namespace/dev created
namespace/prod created
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           27s
frontend-pod   1/1     Running   0           27s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           34s
frontend-pod   1/1     Running   0           34s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backup-job-29468510-9f9cj            0/1     Completed 0           10m
backup-job-29468515-4mwdq            0/1     Completed 0           5m16s
backup-job-29468520-bg5gn            0/1     Completed 0           16s
my-cron-29468510-g7j4f               0/1     Completed 0           10m
my-cron-29468515-mtcf9               0/1     Completed 0           5m16s
my-cron-29468520-78dvt               0/1     Completed 0           16s
nginx-deployment-6f9664446b-n6d2r    1/1     Running   1 (12m ago) 31m
nginx-deployment-6f9664446b-z2vs6    1/1     Running   1 (12m ago) 31m
nodejs-pod                           1/1     Running   7 (12m ago) 31d
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$
```

Step 6: Filter Pods using Labels

List frontend pods in dev: `kubectl get pods -n dev -l app=frontend`

List backend pods in prod: `kubectl get pods -n prod -l app=backend`

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           97s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0          106s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prog9$
```